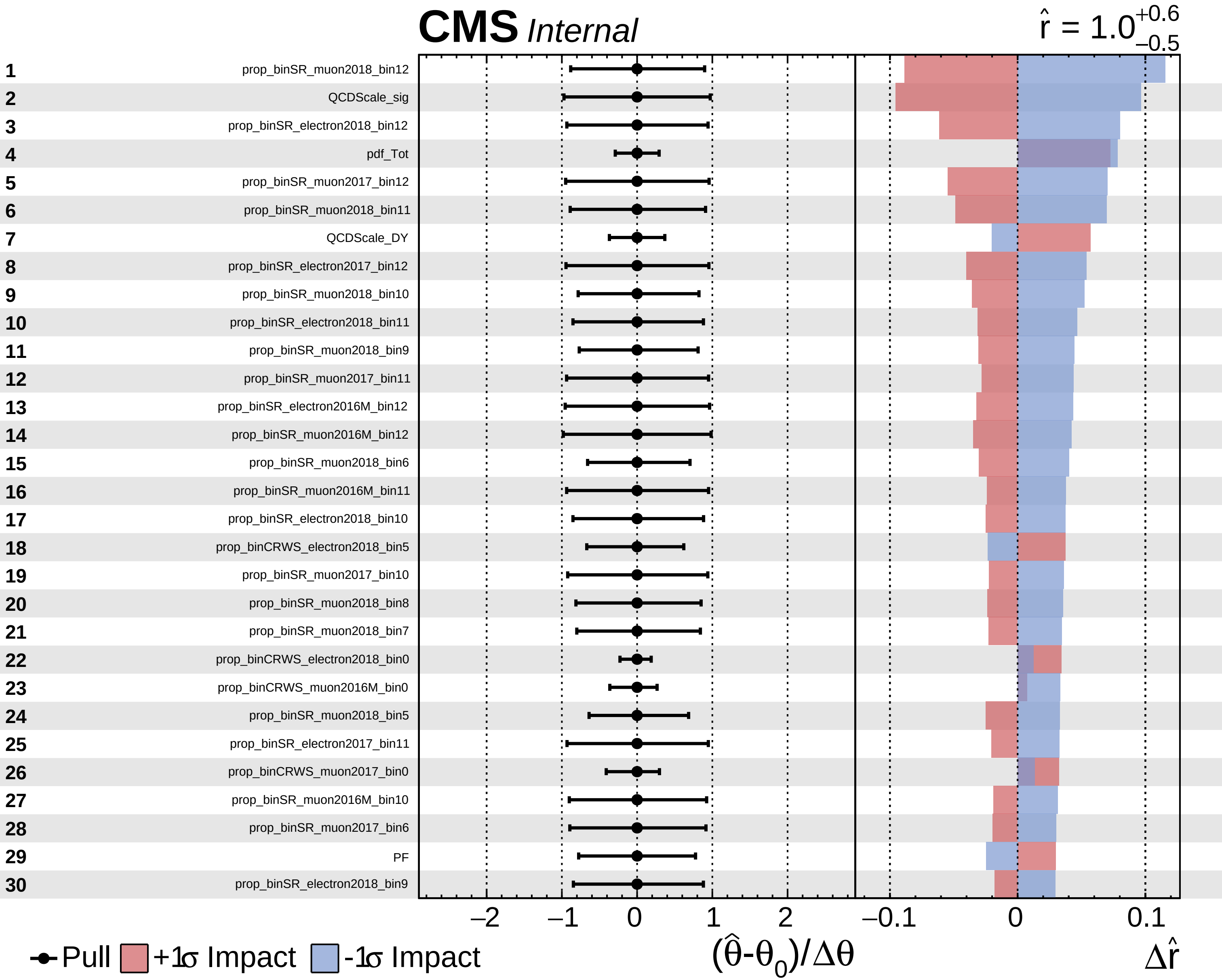
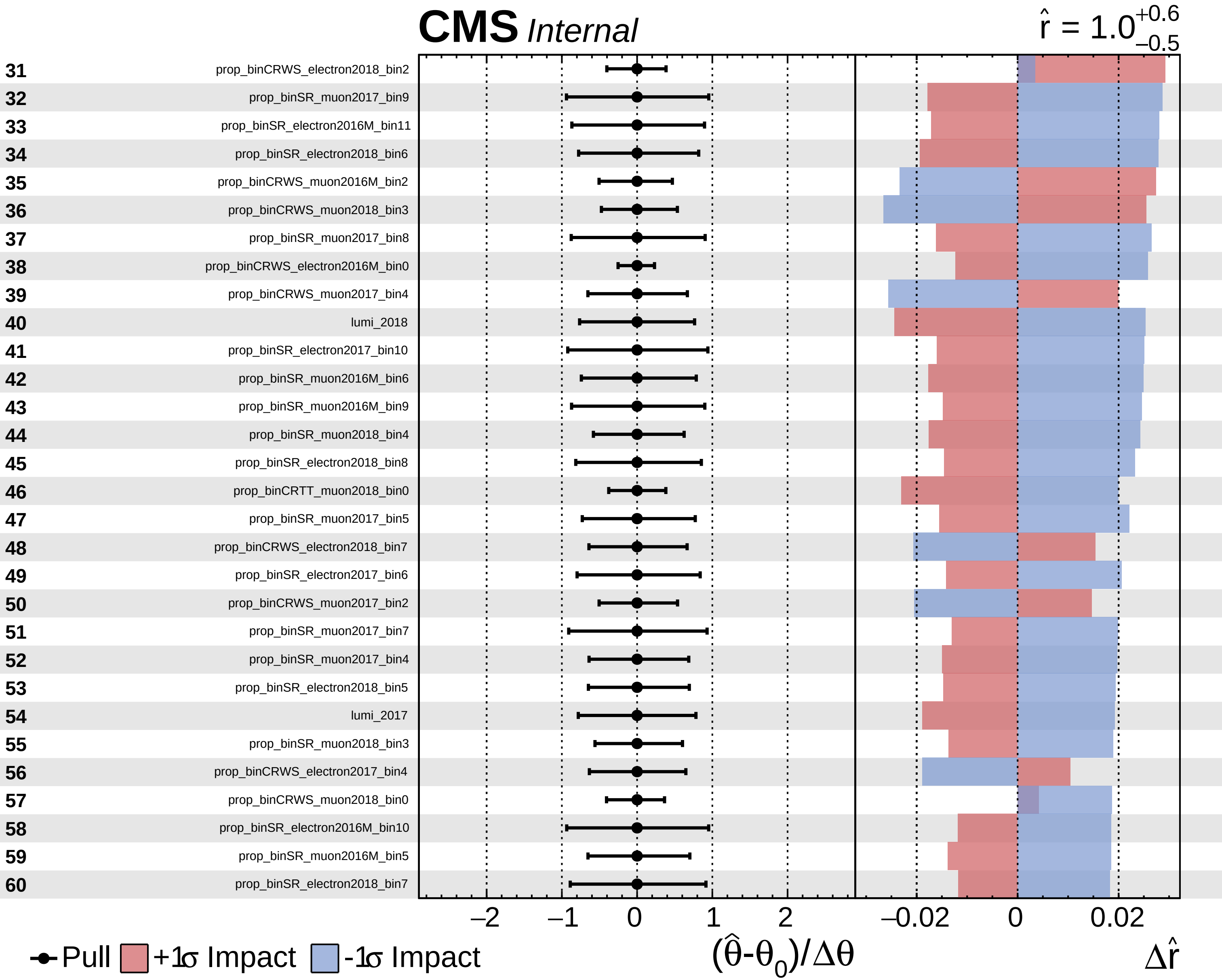
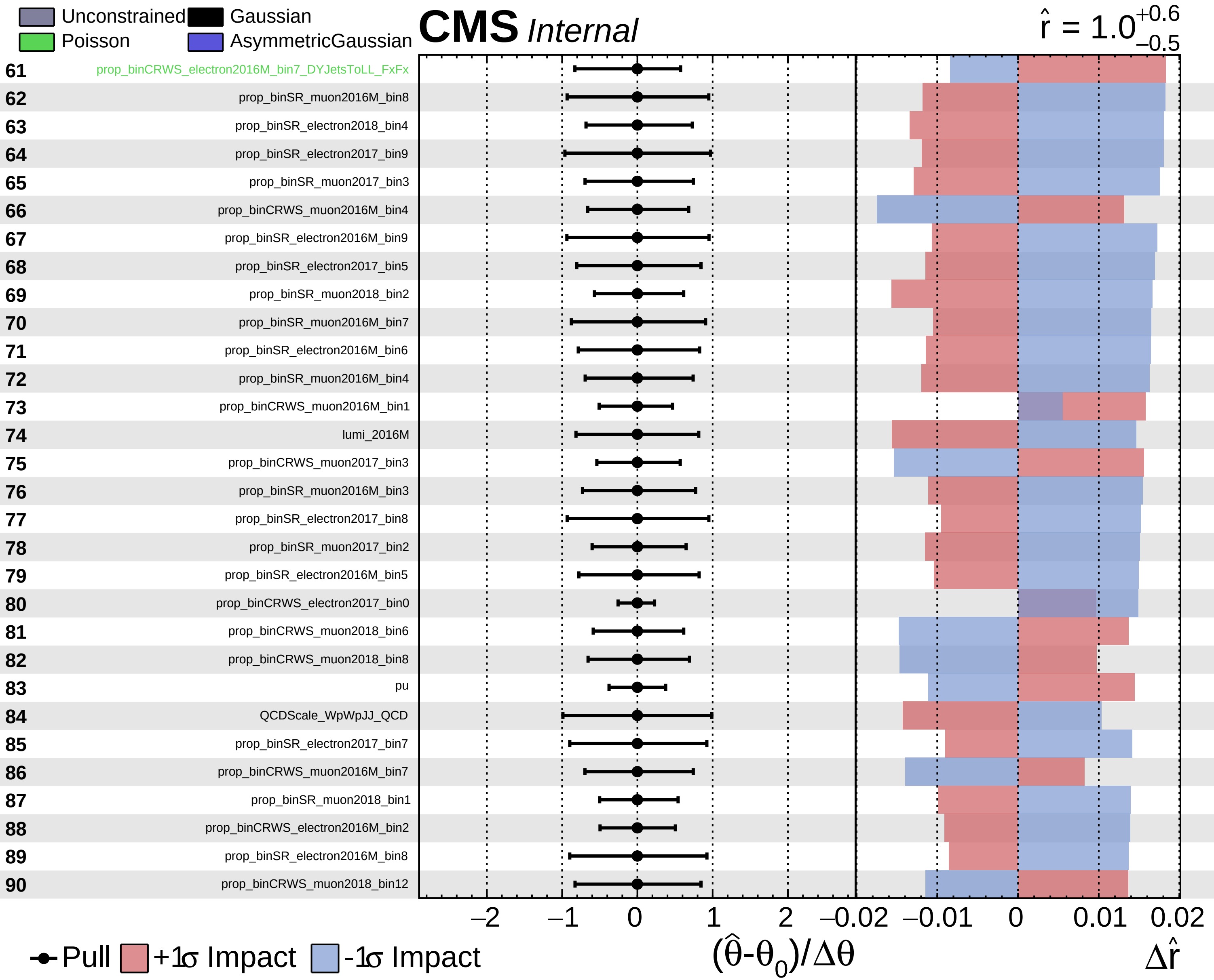


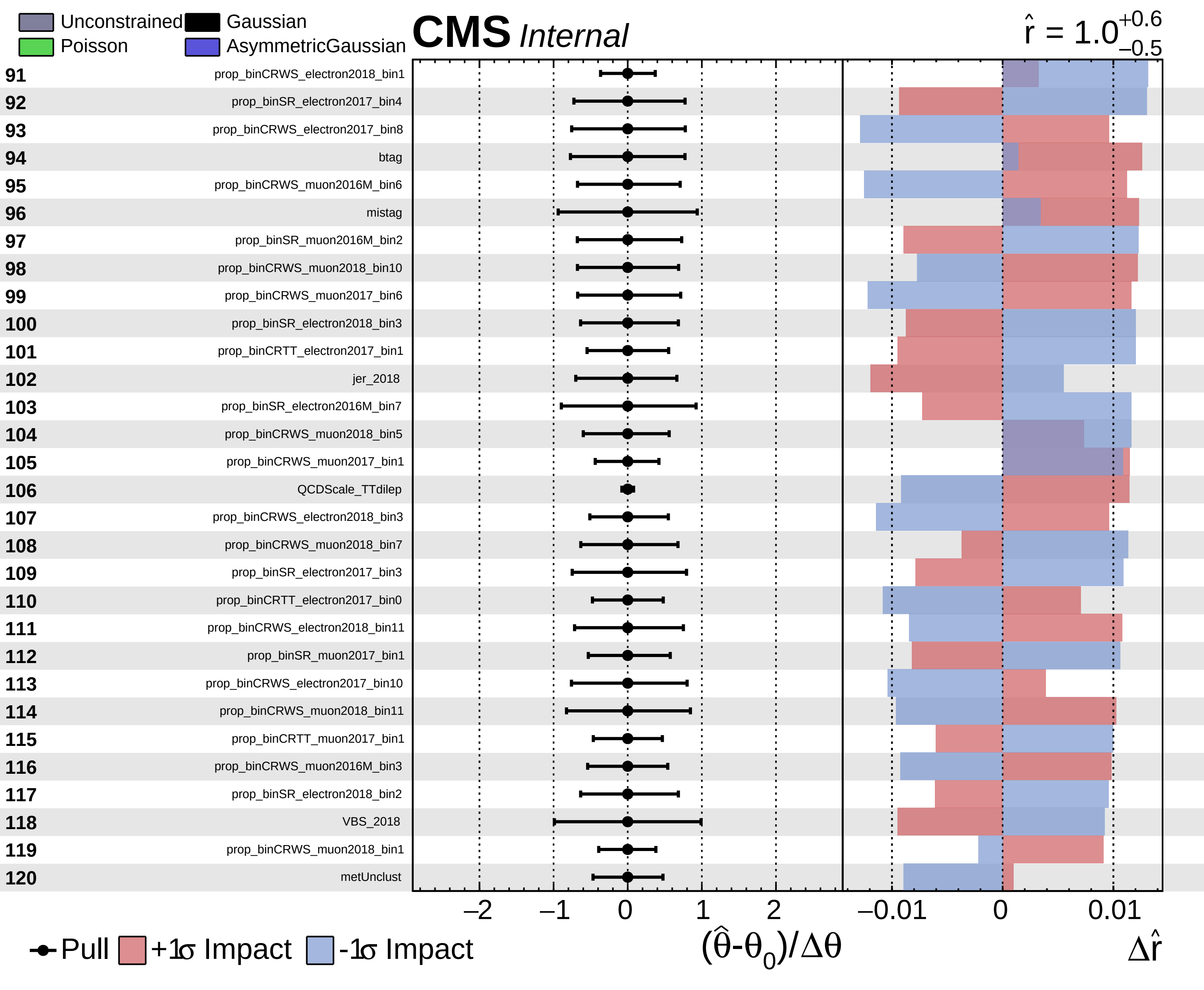
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
1 prop_binSR_muon2018_bin12
2 QCDScale_sig
3 prop_binSR_electron2018_bin12
4 pdf_Tot
5 prop_binSR_muon2017_bin12
6 prop_binSR_muon2018_bin11
7 QCDScale_DY
8 prop_binSR_electron2017_bin12
9 prop_binSR_muon2018_bin10
10 prop_binSR_electron2018_bin11
11 prop_binSR_muon2018_bin9
12 prop_binSR_muon2017_bin11
13 prop_binSR_electron2016M_bin12
14 prop_binSR_muon2016M_bin12
15 prop_binSR_muon2018_bin6
16 prop_binSR_muon2016M_bin11
17 prop_binSR_electron2018_bin10
18 prop_binCRWS_electron2018_bin5
19 prop_binSR_muon2017_bin10
20 prop_binSR_muon2018_bin8
21 prop_binSR_muon2018_bin7
22 prop_binCRWS_electron2018_bin0
23 prop_binCRWS_muon2016M_bin0
24 prop_binSR_muon2018_bin5
25 prop_binSR_electron2017_bin11
26 prop_binCRWS_muon2017_bin0
27 prop_binSR_muon2016M_bin10
28 prop_binSR_muon2017_bin6
29 PF
30 prop_binSR_electron2018_bin9
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

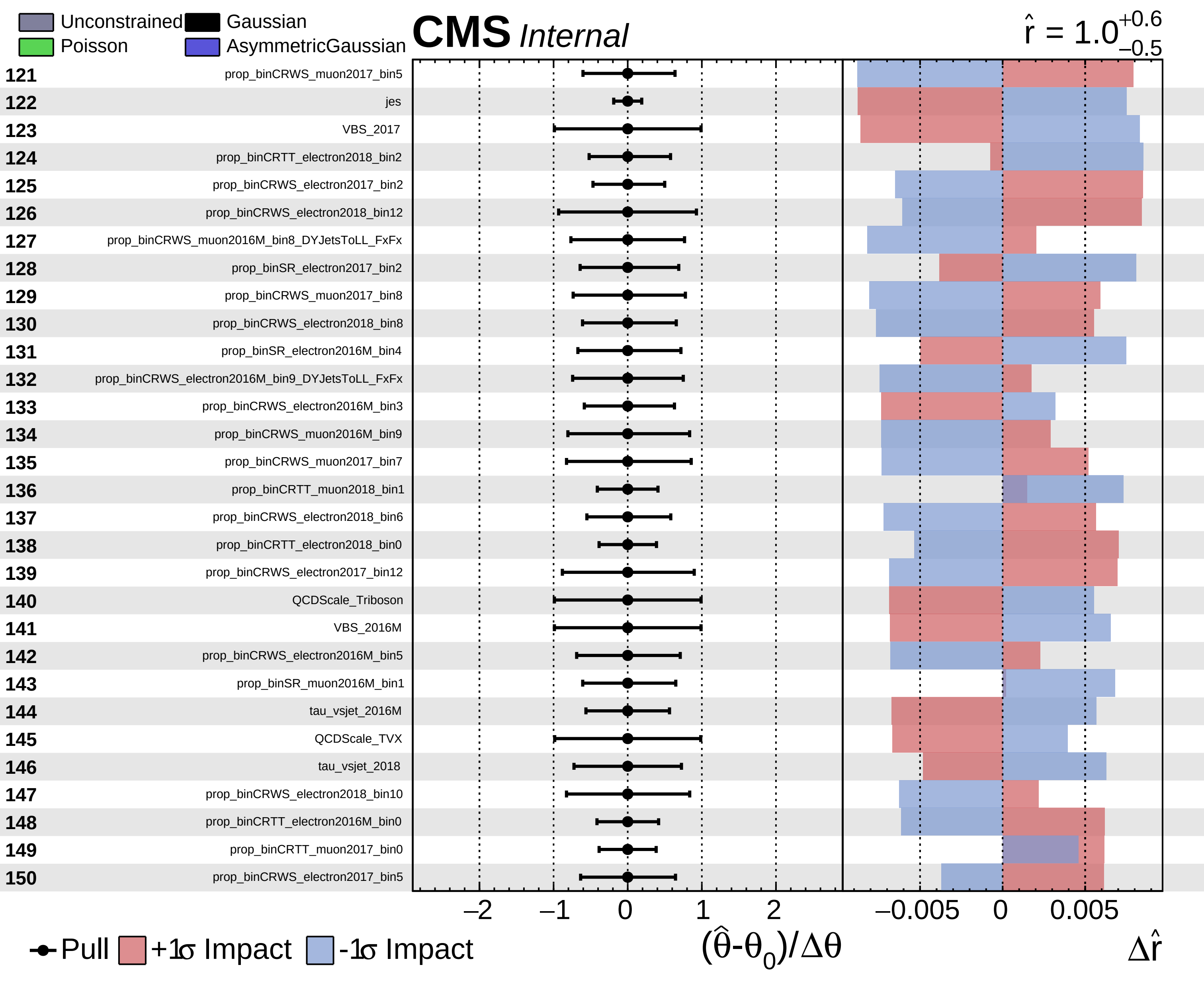


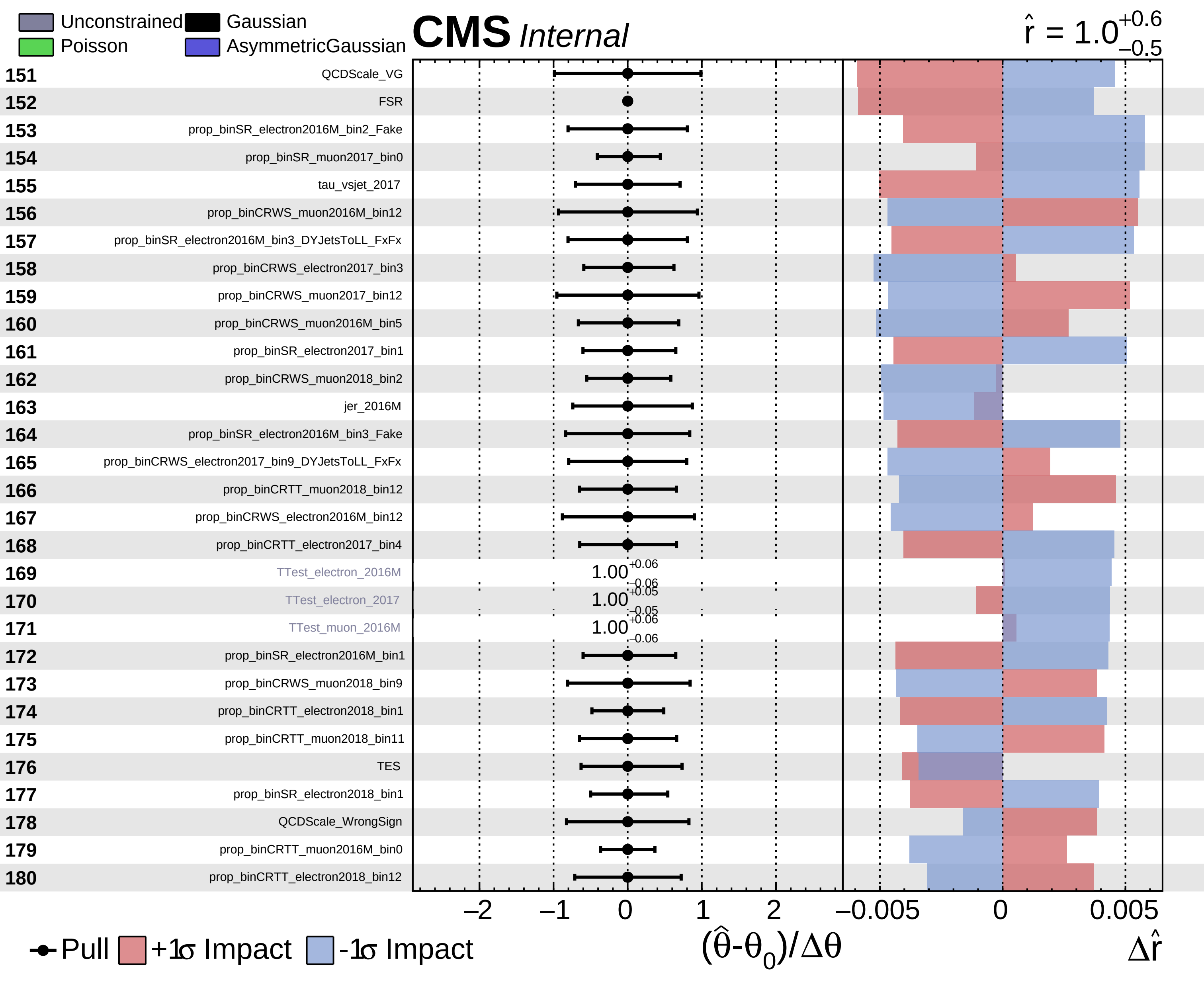
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
31 prop_binCRWS_electron2018_bin2
32 prop_binSR_muon2017_bin9
33 prop_binSR_electron2016M_bin11
34 prop_binSR_electron2018_bin6
35 prop_binCRWS_muon2016M_bin2
36 prop_binCRWS_muon2018_bin3
37 prop_binSR_muon2017_bin8
38 prop_binCRWS_electron2016M_bin0
39 prop_binCRWS_muon2017_bin4
40 lumi_2018
41 prop_binSR_electron2017_bin10
42 prop_binSR_muon2016M_bin6
43 prop_binSR_muon2016M_bin9
44 prop_binSR_muon2018_bin4
45 prop_binSR_electron2018_bin8
46 prop_binCRTT_muon2018_bin0
47 prop_binSR_muon2017_bin5
48 prop_binCRWS_electron2018_bin7
49 prop_binSR_electron2017_bin6
50 prop_binCRWS_muon2017_bin2
51 prop_binSR_muon2017_bin7
52 prop_binSR_muon2017_bin4
53 prop_binSR_electron2018_bin5
54 lumi_2017
55 prop_binSR_muon2018_bin3
56 prop_binCRWS_electron2017_bin4
57 prop_binCRWS_muon2018_bin0
58 prop_binSR_electron2016M_bin10
59 prop_binSR_muon2016M_bin5
60 prop_binSR_electron2018_bin7
−2 −1 0 1 2
−0.02 0 0.02
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



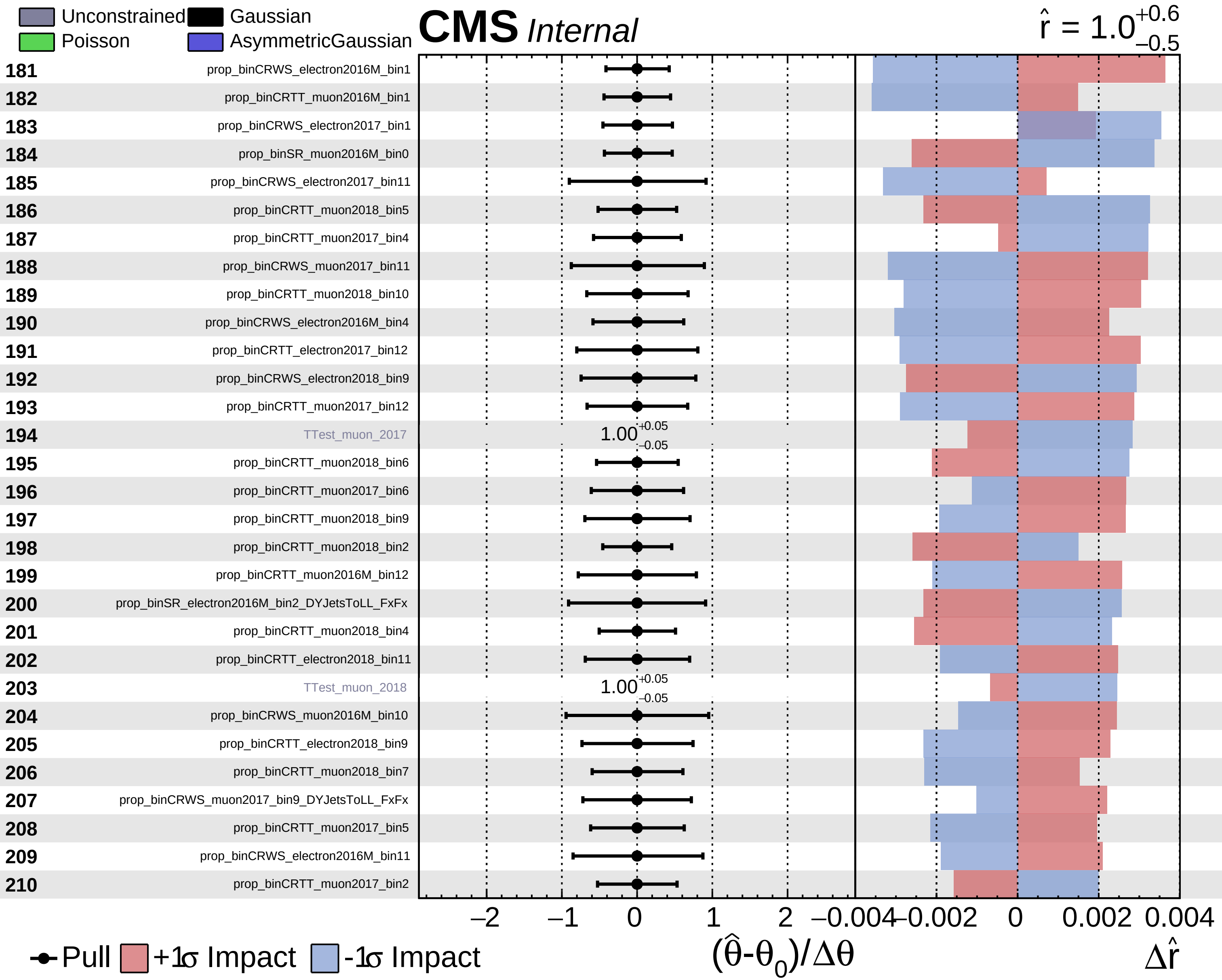
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
61 prop_binCRWS_electron2016M_bin7_DYJetsToLL_FxFx
62 prop_binSR_muon2016M_bin8
63 prop_binSR_electron2018_bin4
64 prop_binSR_electron2017_bin9
65 prop_binSR_muon2017_bin3
66 prop_binCRWS_muon2016M_bin4
67 prop_binSR_electron2016M_bin9
68 prop_binSR_electron2017_bin5
69 prop_binSR_muon2018_bin2
70 prop_binSR_muon2016M_bin7
71 prop_binSR_electron2016M_bin6
72 prop_binSR_muon2016M_bin4
73 prop_binCRWS_muon2016M_bin1
74 lumi_2016M
75 prop_binCRWS_muon2017_bin3
76 prop_binSR_muon2016M_bin3
77 prop_binSR_electron2017_bin8
78 prop_binSR_muon2017_bin2
79 prop_binSR_electron2016M_bin5
80 prop_binCRWS_electron2017_bin0
81 prop_binCRWS_muon2018_bin6
82 prop_binCRWS_muon2018_bin8
83 pu
84 QCDScale_WpWpJJ_QCD
85 prop_binSR_electron2017_bin7
86 prop_binCRWS_muon2016M_bin7
87 prop_binSR_muon2018_bin1
88 prop_binCRWS_electron2016M_bin2
89 prop_binSR_electron2016M_bin8
90 prop_binCRWS_muon2018_bin12
−2
−1
0
1
2
−0.02
−0.01
0
0.01
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

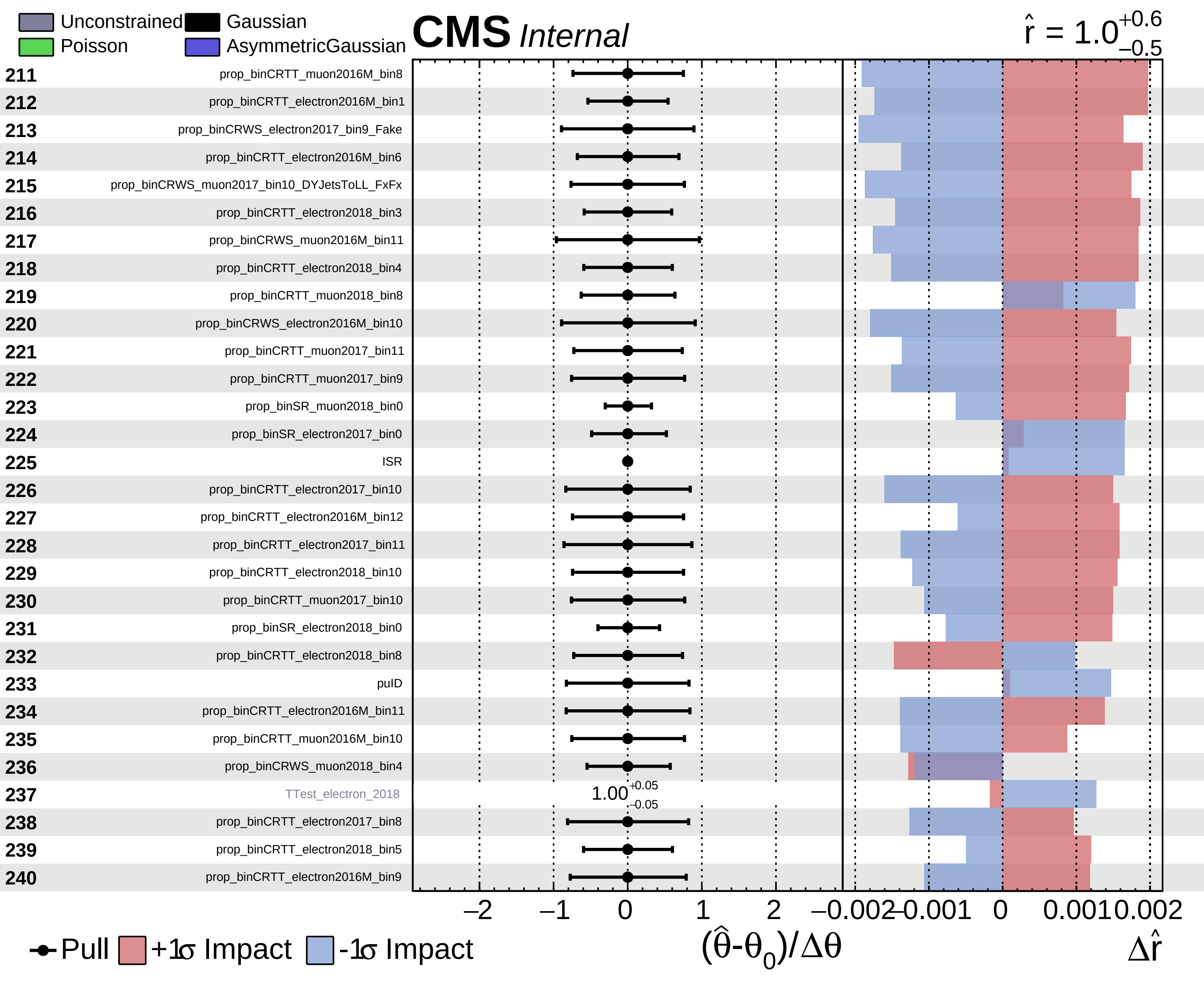

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
91 prop_binCRWS_electron2018_bin1
92 prop_binSR_electron2017_bin4
93 prop_binCRWS_electron2017_bin8
94 btag
95 prop_binCRWS_muon2016M_bin6
96 mistag
97 prop_binSR_muon2016M_bin2
98 prop_binCRWS_muon2018_bin10
99 prop_binCRWS_muon2017_bin6
100 prop_binSR_electron2018_bin3
101 prop_binCRTT_electron2017_bin1
102 jer_2018
103 prop_binSR_electron2016M_bin7
104 prop_binCRWS_muon2018_bin5
105 prop_binCRWS_muon2017_bin1
106 QCDScale_TTdilep
107 prop_binCRWS_electron2018_bin3
108 prop_binCRWS_muon2018_bin7
109 prop_binSR_electron2017_bin3
110 prop_binCRTT_electron2017_bin0
111 prop_binCRWS_electron2018_bin11
112 prop_binSR_muon2017_bin1
113 prop_binCRWS_electron2017_bin10
114 prop_binCRWS_muon2018_bin11
115 prop_binCRTT_muon2017_bin1
116 prop_binCRWS_muon2016M_bin3
117 prop_binSR_electron2018_bin2
118 VBS_2018
119 prop_binCRWS_muon2018_bin1
120 metUnclust
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
121 prop_binCRWS_muon2017_bin5
122 jes
123 VBS_2017
124 prop_binCRTT_electron2018_bin2
125 prop_binCRWS_electron2017_bin2
126 prop_binCRWS_electron2018_bin12
127 prop_binCRWS_muon2016M_bin8_DYJetsToLL_FxFx
128 prop_binSR_electron2017_bin2
129 prop_binCRWS_muon2017_bin8
130 prop_binCRWS_electron2018_bin8
131 prop_binSR_electron2016M_bin4
132 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
133 prop_binCRWS_electron2016M_bin3
134 prop_binCRWS_muon2016M_bin9
135 prop_binCRWS_muon2017_bin7
136 prop_binCRTT_muon2018_bin1
137 prop_binCRWS_electron2018_bin6
138 prop_binCRTT_electron2018_bin0
139 prop_binCRWS_electron2017_bin12
140 QCDScale_Triboson
141 VBS_2016M
142 prop_binCRWS_electron2016M_bin5
143 prop_binSR_muon2016M_bin1
144 tau_vsjet_2016M
145 QCDScale_TVX
146 tau_vsjet_2018
147 prop_binCRWS_electron2018_bin10
148 prop_binCRTT_electron2016M_bin0
149 prop_binCRTT_muon2017_bin0
150 prop_binCRWS_electron2017_bin5
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

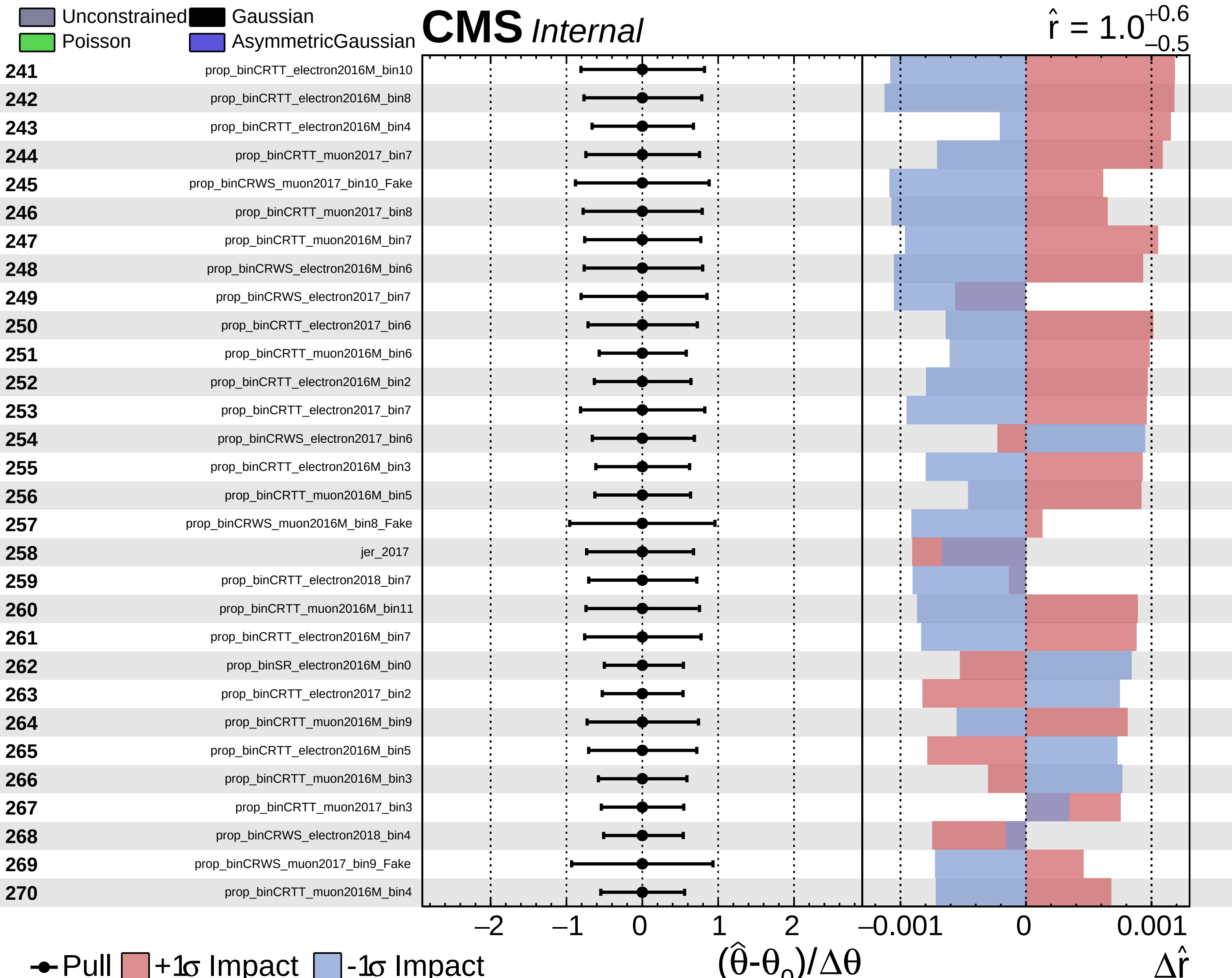


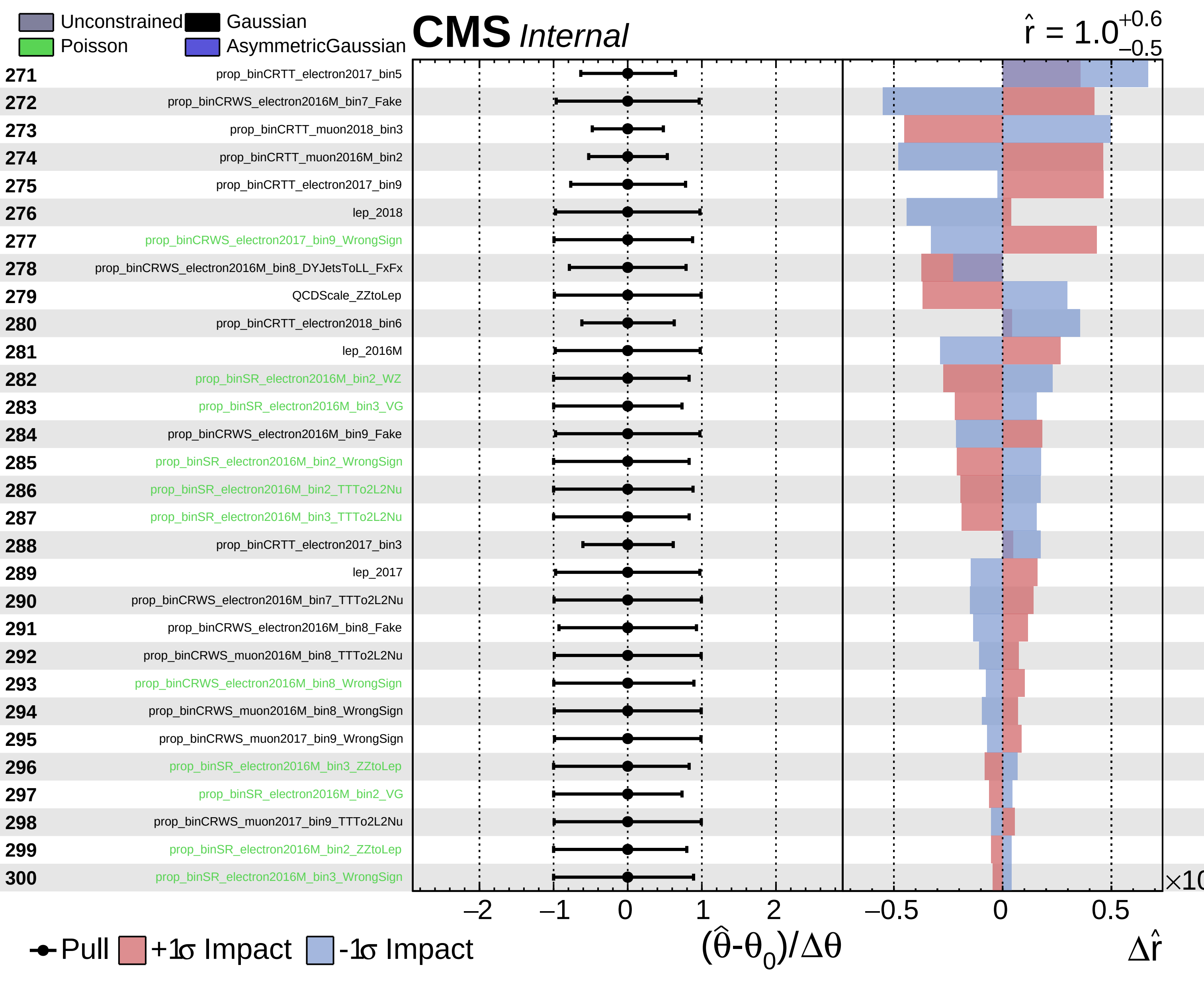
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.6 −0.5
151 QCDScale_VG
152 FSR
153 prop_binSR_electron2016M_bin2_Fake
154 prop_binSR_muon2017_bin0
155 tau_vsjet_2017
156 prop_binCRWS_muon2016M_bin12
157 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
158 prop_binCRWS_electron2017_bin3
159 prop_binCRWS_muon2017_bin12
160 prop_binCRWS_muon2016M_bin5
161 prop_binSR_electron2017_bin1
162 prop_binCRWS_muon2018_bin2
163 jer_2016M
164 prop_binSR_electron2016M_bin3_Fake
165 prop_binCRWS_electron2017_bin9_DYJetsToLL_FxFx
166 prop_binCRTT_muon2018_bin12
167 prop_binCRWS_electron2016M_bin12
168 prop_binCRTT_electron2017_bin4
169 TTest_electron_2016M 1.00 +0.06 −0.06
170 TTest_electron_2017 1.00 +0.05 −0.05
171 TTest_muon_2016M 1.00 +0.06 −0.06
172 prop_binSR_electron2016M_bin1
173 prop_binCRWS_muon2018_bin9
174 prop_binCRTT_electron2018_bin1
175 prop_binCRTT_muon2018_bin11
176 TES
177 prop_binSR_electron2018_bin1
178 QCDScale_WrongSign
179 prop_binCRTT_muon2016M_bin0
180 prop_binCRTT_electron2018_bin12
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂

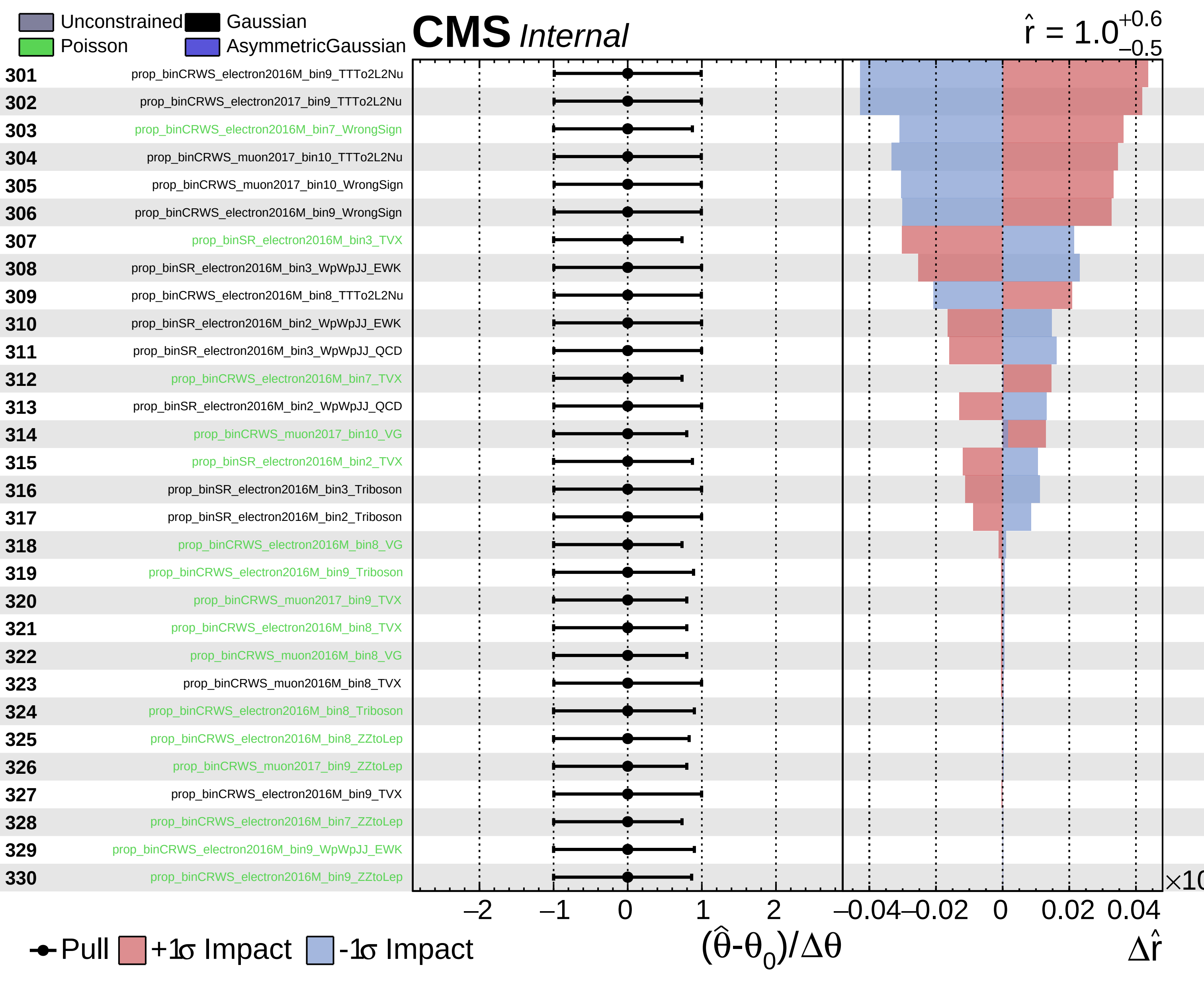

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
181 prop_binCRWS_electron2016M_bin1
182 prop_binCRTT_muon2016M_bin1
183 prop_binCRWS_electron2017_bin1
184 prop_binSR_muon2016M_bin0
185 prop_binCRWS_electron2017_bin11
186 prop_binCRTT_muon2018_bin5
187 prop_binCRTT_muon2017_bin4
188 prop_binCRWS_muon2017_bin11
189 prop_binCRTT_muon2018_bin10
190 prop_binCRWS_electron2016M_bin4
191 prop_binCRTT_electron2017_bin12
192 prop_binCRWS_electron2018_bin9
193 prop_binCRTT_muon2017_bin12
194 TTest_muon_2017 $1.00^{+0.05}_{-0.05}$
195 prop_binCRTT_muon2018_bin6
196 prop_binCRTT_muon2017_bin6
197 prop_binCRTT_muon2018_bin9
198 prop_binCRTT_muon2018_bin2
199 prop_binCRTT_muon2016M_bin12
200 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
201 prop_binCRTT_muon2018_bin4
202 prop_binCRTT_electron2018_bin11
203 TTest_muon_2018 $1.00^{+0.05}_{-0.05}$
204 prop_binCRWS_muon2016M_bin10
205 prop_binCRTT_electron2018_bin9
206 prop_binCRTT_muon2018_bin7
207 prop_binCRWS_muon2017_bin9_DYJetsToLL_FxFx
208 prop_binCRTT_muon2017_bin5
209 prop_binCRWS_electron2016M_bin11
210 prop_binCRTT_muon2017_bin2
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



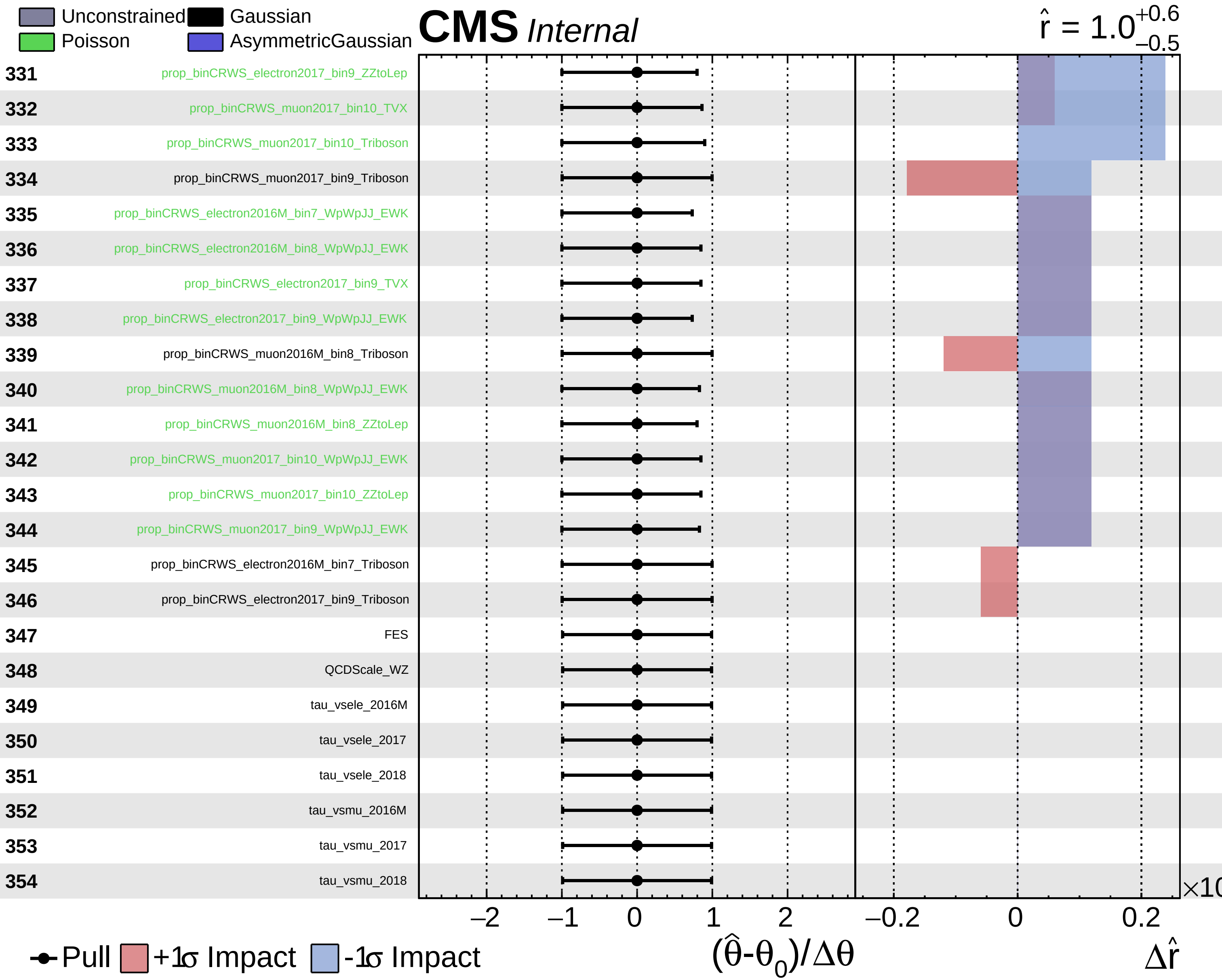
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
211 prop_binCRTT_muon2016M_bin8
212 prop_binCRTT_electron2016M_bin1
213 prop_binCRWS_electron2017_bin9_Fake
214 prop_binCRTT_electron2016M_bin6
215 prop_binCRWS_muon2017_bin10_DYJetsToLL_FxFx
216 prop_binCRTT_electron2018_bin3
217 prop_binCRWS_muon2016M_bin11
218 prop_binCRTT_electron2018_bin4
219 prop_binCRTT_muon2018_bin8
220 prop_binCRWS_electron2016M_bin10
221 prop_binCRTT_muon2017_bin11
222 prop_binCRTT_muon2017_bin9
223 prop_binSR_muon2018_bin0
224 prop_binSR_electron2017_bin0
225 ISR
226 prop_binCRTT_electron2017_bin10
227 prop_binCRTT_electron2016M_bin12
228 prop_binCRTT_electron2017_bin11
229 prop_binCRTT_electron2018_bin10
230 prop_binCRTT_muon2017_bin10
231 prop_binSR_electron2018_bin0
232 prop_binCRTT_electron2018_bin8
233 puID
234 prop_binCRTT_electron2016M_bin11
235 prop_binCRTT_muon2016M_bin10
236 prop_binCRWS_muon2018_bin4
237 TTest_electron_2018 $1.00^{+0.05}_{-0.05}$
238 prop_binCRTT_electron2017_bin8
239 prop_binCRTT_electron2018_bin5
240 prop_binCRTT_electron2016M_bin9
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
241 prop_binCRTT_electron2016M_bin10
242 prop_binCRTT_electron2016M_bin8
243 prop_binCRTT_electron2016M_bin4
244 prop_binCRTT_muon2017_bin7
245 prop_binCRWS_muon2017_bin10_Fake
246 prop_binCRTT_muon2017_bin8
247 prop_binCRTT_muon2016M_bin7
248 prop_binCRWS_electron2016M_bin6
249 prop_binCRWS_electron2017_bin7
250 prop_binCRTT_electron2017_bin6
251 prop_binCRTT_muon2016M_bin6
252 prop_binCRTT_electron2016M_bin2
253 prop_binCRTT_electron2017_bin7
254 prop_binCRWS_electron2017_bin6
255 prop_binCRTT_electron2016M_bin3
256 prop_binCRTT_muon2016M_bin5
257 prop_binCRWS_muon2016M_bin8_Fake
258 jer_2017
259 prop_binCRTT_electron2018_bin7
260 prop_binCRTT_muon2016M_bin11
261 prop_binCRTT_electron2016M_bin7
262 prop_binSR_electron2016M_bin0
263 prop_binCRTT_electron2017_bin2
264 prop_binCRTT_muon2016M_bin9
265 prop_binCRTT_electron2016M_bin5
266 prop_binCRTT_muon2016M_bin3
267 prop_binCRTT_muon2017_bin3
268 prop_binCRWS_electron2018_bin4
269 prop_binCRWS_muon2017_bin9_Fake
270 prop_binCRTT_muon2016M_bin4
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
271 prop_binCRTT_electron2017_bin5
272 prop_binCRWS_electron2016M_bin7_Fake
273 prop_binCRTT_muon2018_bin3
274 prop_binCRTT_muon2016M_bin2
275 prop_binCRTT_electron2017_bin9
276 lep_2018
277 prop_binCRWS_electron2017_bin9_WrongSign
278 prop_binCRWS_electron2016M_bin8_DYJetsToLL_FxFx
279 QCDScale_ZZtoLep
280 prop_binCRTT_electron2018_bin6
281 lep_2016M
282 prop_binSR_electron2016M_bin2_WZ
283 prop_binSR_electron2016M_bin3_VG
284 prop_binCRWS_electron2016M_bin9_Fake
285 prop_binSR_electron2016M_bin2_WrongSign
286 prop_binSR_electron2016M_bin2_TTTo2L2Nu
287 prop_binSR_electron2016M_bin3_TTTo2L2Nu
288 prop_binCRTT_electron2017_bin3
289 lep_2017
290 prop_binCRWS_electron2016M_bin7_TTTo2L2Nu
291 prop_binCRWS_electron2016M_bin8_Fake
292 prop_binCRWS_muon2016M_bin8_TTTo2L2Nu
293 prop_binCRWS_electron2016M_bin8_WrongSign
294 prop_binCRWS_muon2016M_bin8_WrongSign
295 prop_binCRWS_muon2017_bin9_WrongSign
296 prop_binSR_electron2016M_bin3_ZZtoLep
297 prop_binSR_electron2016M_bin2_VG
298 prop_binCRWS_muon2017_bin9_TTTo2L2Nu
299 prop_binSR_electron2016M_bin2_ZZtoLep
300 prop_binSR_electron2016M_bin3_WrongSign
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
301 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
302 prop_binCRWS_electron2017_bin9_TTTo2L2Nu
303 prop_binCRWS_electron2016M_bin7_WrongSign
304 prop_binCRWS_muon2017_bin10_TTTo2L2Nu
305 prop_binCRWS_muon2017_bin10_WrongSign
306 prop_binCRWS_electron2016M_bin9_WrongSign
307 prop_binSR_electron2016M_bin3_TVX
308 prop_binSR_electron2016M_bin3_WpWpJJ_EWK
309 prop_binCRWS_electron2016M_bin8_TTTo2L2Nu
310 prop_binSR_electron2016M_bin2_WpWpJJ_EWK
311 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
312 prop_binCRWS_electron2016M_bin7_TVX
313 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
314 prop_binCRWS_muon2017_bin10_VG
315 prop_binSR_electron2016M_bin2_TVX
316 prop_binSR_electron2016M_bin3_Triboson
317 prop_binSR_electron2016M_bin2_Triboson
318 prop_binCRWS_electron2016M_bin8_VG
319 prop_binCRWS_electron2016M_bin9_Triboson
320 prop_binCRWS_muon2017_bin9_TVX
321 prop_binCRWS_electron2016M_bin8_TVX
322 prop_binCRWS_muon2016M_bin8_VG
323 prop_binCRWS_muon2016M_bin8_TVX
324 prop_binCRWS_electron2016M_bin8_Triboson
325 prop_binCRWS_electron2016M_bin8_ZZtoLep
326 prop_binCRWS_muon2017_bin9_ZZtoLep
327 prop_binCRWS_electron2016M_bin9_TVX
328 prop_binCRWS_electron2016M_bin7_ZZtoLep
329 prop_binCRWS_electron2016M_bin9_WpWpJJ_EWK
330 prop_binCRWS_electron2016M_bin9_ZZtoLep
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.6}_{-0.5}$
331 prop_binCRWS_electron2017_bin9_ZZtoLep
332 prop_binCRWS_muon2017_bin10_TVX
333 prop_binCRWS_muon2017_bin10_Triboson
334 prop_binCRWS_muon2017_bin9_Triboson
335 prop_binCRWS_electron2016M_bin7_WpWpJJ_EWK
336 prop_binCRWS_electron2016M_bin8_WpWpJJ_EWK
337 prop_binCRWS_electron2017_bin9_TVX
338 prop_binCRWS_electron2017_bin9_WpWpJJ_EWK
339 prop_binCRWS_muon2016M_bin8_Triboson
340 prop_binCRWS_muon2016M_bin8_WpWpJJ_EWK
341 prop_binCRWS_muon2016M_bin8_ZZtoLep
342 prop_binCRWS_muon2017_bin10_WpWpJJ_EWK
343 prop_binCRWS_muon2017_bin10_ZZtoLep
344 prop_binCRWS_muon2017_bin9_WpWpJJ_EWK
345 prop_binCRWS_electron2016M_bin7_Triboson
346 prop_binCRWS_electron2017_bin9_Triboson
347 FES
348 QCDScale_WZ
349 tau_vsele_2016M
350 tau_vsele_2017
351 tau_vsele_2018
352 tau_vsmu_2016M
353 tau_vsmu_2017
354 tau_vsmu_2018
−2
−1
0
1
2
−0.2
0
0.2
$\times 10$
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$